QPCODE: 25900783 Scheme of Evaluation

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025 FIRST SEMESTER

Core Course (CC) - MG1CCRBCA100 - DIGITAL FUNDAMENTALS

Duration: 2 Hours **Maximum Marks: 70**

Part A-Very Short Answer Questions, Answer All Questions, Each Question carries 2 marks

1. Explain the process of converting a decimal number to binary.-repeatedly divide the number by 2 and track the remainders. The binary number is then formed by reading the remainders in reverse order. [U] / [CO1]

2. Explain the concept of overflow in binary addition. [U] / [CO1]

   Overflow in binary addition happens when the result of an addition is too large to fit in the number of bits available. In other words, the answer exceeds the maximum value that can be represented with the given number of bits.

3. Convert the expression into sum-of-product (SOP) form-(A+B)(A+B+C). [A] / [CO2]

   Simplified sum of product form is A+B. Give upmarks to intermediate results in SOP.

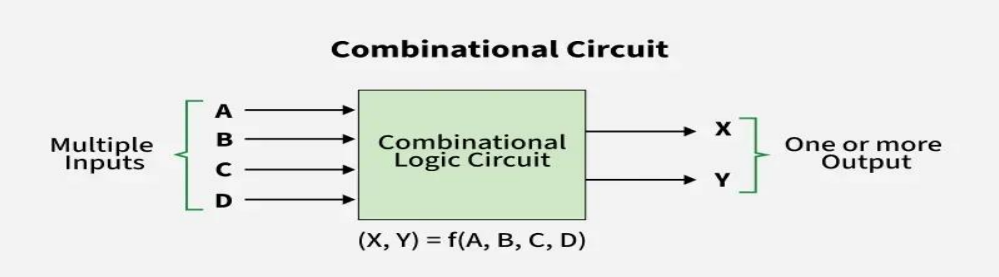
4. Simplify the expression f = A + B[AC + (B + C′) D] using basic laws of Boolean algebra. [A] / [CO2]

   Simplified expression is A+BD. Give marks up to 1.5 to simplification using basic laws and intermediate results

5. What is the use of a de-multiplexer? [K] / [CO3]

   A demultiplexer is used to route a single input signal to one of several output lines based on select signals. It is widely used in data routing, communication systems, and converting serial data to parallel.

6. Draw the block diagram of a combinational logic circuit. [U] / [CO3]



7. Define sequential circuits and explain their significance. [U] / [CO4]

   Sequential circuits are digital circuits that use memory elements to store previous states, allowing their outputs to depend on both current inputs and past inputs. This characteristic makes them significant for implementing memory, state machines, counters, and other systems that require processing information over time

8. Explain the concept of toggling in a JK flip-flop. [U] / [CO4]

   A JK flip-flop toggles when both inputs, J and K, are set to 1. This means that on the next clock pulse, the output Q will flip to 1if it was 0, or to 0 if it was 1

   . This behavior makes the JK flip-flop useful for applications like counters

9. Convert the binary number 101011 to octal. Ans: Octal 53. Two marks for correct answer with steps [A] / [CO1]

10. What is the primary difference between a half adder and a full adder? [U] / [CO3]

    A half adder adds two single-bit binary numbers, while a full adder adds three single-bit binary numbers, including a carry-in from a previous operation.

[10x2 = 20]

Part B, Answer any **5** out of **7** Questions, Each Question carries **6** marks

11. Convert the decimal number 357 to its BCD representation using the 8421 code. Show each step of the conversion process. [A] / [CO1]

    Answer : 0011 0101 0111 (3 marks explanation or steps 3 marks)

12. Simplify the expression F(x,y,z)=∑(0,6) and implement with NOR gate. [A] / [CO2]

    Simplification using K-map 3 marks, implementation using NOR gate 3 marks

13. Explain how a decoder is used for memory address selection in computer systems. - by converting a binary memory address from the CPU into a single active output line that enables a particular memory chip or a specific word line within a chip. It takes n input lines representing the binary address and activates one of its 2^n output lines, where only the selected line is active (e.g., high) at any given time. This ensures that data is read from or written to only the correct memory cell. [U] / [CO3]

14. Discuss the characteristics and use of a D flip-flop. – characteristics table or discussion -4 marks. Use of D fip-flop 2 marks [U] / [CO4]

15. Compare and contrast the binary, octal, and hexadecimal number systems. Discuss the benefits of each in computing. – Comparison 4 marks benefits explanation 2 marks [U] / [CO1]

16. Apply De Morgan's theorems to each of the following expressions (i) [(A + B + C) D]′ (ii) (ABC + DEF)′ (iii) (AB′+ C′D + EF)′. – 2 marks each [A] / [CO2]

17. Compare the behaviour of positive-edge-triggered and negative-edge- triggered flip-flops based on a timing diagram. –2 marks each for definition 2 marks for comparison [An] / [CO4]

[5x6 = 30]

Part C- Answer any **2** out of **3** Questions, Each Question carries **10** marks

18. Simplify the Boolean function using K-Map: F(w,x,y,z)=Σ(1,3,4,5,6,9,12,13). – k-map 4 marks, simplification 4 marks, Simplified expression 2 marks [A] / [CO2]

19. Design an 8×1 multiplexer using 4×1 and 2×1 multiplexer. – Design and diagram 6 marks, explanation 4 marks [A] / [CO3]

20. Describe the operation of a 4-bit serial-in parallel-out (SIPO) shift register. Include a timing diagram to show how data is shifted in and read out. Description with block diagram 6 marks, timing diagram and explanation 4 marks [U] / [CO4]

[2x10 = 20]

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